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OPEN-SOURCE AUDIT PROJECT – GIT

1. Introduction

Open-Source Software (OSS) plays a major role in today's technology landscape. It gives users the freedom to access source code, understand how software works, make changes, and share it with others. This approach promotes transparency, collaboration, continuous learning, and innovation.

In this project, Git has been analyzed as an example of open-source software. The audit covers its background, key features, licensing, open-source principles, and real-world applications. It also includes simple shell scripts to demonstrate basic Linux command-line skills.

2. About the Selected Software

The software selected for this project is Git. Git is a distributed version control system developed by Linus Torvalds in 2005. It was originally created to manage the Linux kernel, but it has since grown into one of the most widely used tools in software development.

Git allows developers to track changes in files and source code over time. It is especially useful in team environments, where multiple developers can work on the same project independently and later merge their contributions efficiently.

3. Main Features of Git

Git provides several powerful features that make it highly popular:

- **Distributed Version Control:** Every user has a full copy of the repository, including its history.
- **Branching and Merging:** Developers can work on separate branches and merge them when ready.
- **High Performance:** Git is fast and efficient, even with large projects.
- **Data Integrity:** It uses hashing to ensure data accuracy and consistency.
- **Collaboration Support:** Git integrates with platforms like GitHub, GitLab, and Bitbucket to enable teamwork.

4. License Analysis

Git is released under the GNU General Public License (GPL), a widely used open-source license that protects user freedoms.

The GPL allows users to:

- Use the software for any purpose
- Study how it works
- Modify the source code
- Share copies
- Distribute modified versions

A key principle of the GPL is that any distributed modifications must also remain open, ensuring continued software freedom.

5. Open-Source Philosophy
Git strongly reflects the core values of open-source software:

- **Transparency:** Anyone can view and inspect the code.
- **Collaboration:** Developers worldwide can contribute improvements.
- **Freedom:** Users can use, modify, and share the software freely.
- **Community-driven Development:** Ongoing improvements come from a global community.

These principles make Git more than just a tool—it represents how open-source drives innovation and learning.

6. Real-World Impact
Git has had a significant impact on the software industry. It is used by individuals, startups, universities, and large organizations alike. It also powers platforms like GitHub, which host millions of projects.

With Git, developers can:

- Collaborate more effectively
- Track detailed project history
- Build software more efficiently
- Manage code changes in an organized way
- Support large open-source communities

Because of these advantages, Git has become a standard in software development.

7. Practical Implementation
In this project, basic Git commands were practiced:

```
git init
git add .
git commit -m "Initial commit"
```

These commands are used to:

- Initialize a new repository
- Add files for tracking
- Save changes with a commit message

This hands-on practice helped in understanding how Git works in real-world scenarios.

8. Shell Scripts Implementation

To support the practical section, five shell scripts were created and tested using Git Bash:

- **File Counter Script:** Counts files in a directory
- **Backup Script:** Creates backups with timestamps
- **System Information Script:** Displays user and system details
- **File Check Script:** Verifies if a file exists
- **Calculator Script:** Performs basic arithmetic

These scripts demonstrate fundamental concepts such as variables, conditions, user input, and automation.

9. Testing and Results

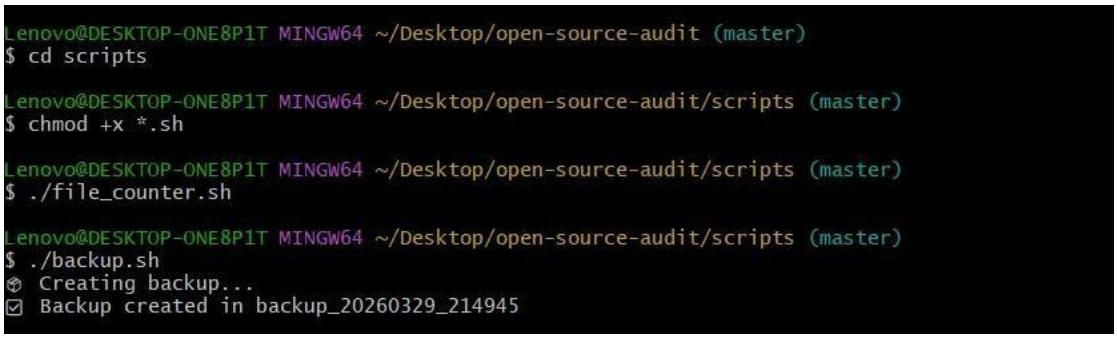
All scripts were tested successfully in Git Bash. Each one produced the expected results without major issues. Screenshots were also taken as proof of successful execution.

10. Conclusion

This project provided valuable insight into open-source software and its importance in modern technology. Studying Git helped build an understanding of its features, licensing, and role in software development.

Additionally, the practical exercises improved skills in Linux commands and shell scripting. Overall, this audit highlights how open-source tools like Git support collaboration, transparency, and innovation in the tech world.

11. Here are some screenshots of it



```
Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ ./system_info.sh
System Information
---------------------
User:
Hostname: DESKTOP-ONE8P1T
Date: Sun Mar 29 21:50:02 IST 2026
Current Directory: /c/Users/Lenovo/Desktop/open-source-audit/scripts
OS: Msys

Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ ./check_file.sh
Enter filename: README.md
✗ File does not exist

Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ ./calculator.sh
Simple Calculator
Enter first number: 12
Enter second number: 12
Choose operation (+ - * /):
/
Result: 1
```

```
Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ ./file_counter.sh
Counting files in current directory...
Total files: 6

Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ ./check_file.sh
Enter filename: ../README.md
☑ File exists

Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ git init
Initialized empty Git repository in C:/Users/Lenovo/Desktop/open-source-audit/scripts/.git/

Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ git add .
warning: in the working copy of 'backup.sh', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'backup_20260329_214945/backup.sh', LF will be replaced by CRLF the nex
t time Git touches it
warning: in the working copy of 'backup_20260329_214945/calculator.sh', LF will be replaced by CRLF the
 next time Git touches it
warning: in the working copy of 'backup_20260329_214945/check_file.sh', LF will be replaced by CRLF the
 next time Git touches it
warning: in the working copy of 'backup_20260329_214945/system_info.sh', LF will be replaced by CRLF th
e next time Git touches it
warning: in the working copy of 'calculator.sh', LF will be replaced by CRLF the next time Git touches
it
warning: in the working copy of 'check_file.sh', LF will be replaced by CRLF the next time Git touches
it
warning: in the working copy of 'file_counter.sh', LF will be replaced by CRLF the next time Git touche
s it
warning: in the working copy of 'system_info.sh', LF will be replaced by CRLF the next time Git touches
 it

Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ git commit -m "Intial commit"
[master (root-commit) 6553d8c] Intial commit
 10 files changed, 94 insertions(+)
 create mode 100644 backup.sh
 create mode 100644 backup_20260329_214945/backup.sh
 create mode 100644 backup_20260329_214945/calculator.sh
 create mode 100644 backup_20260329_214945/check_file.sh
 create mode 100644 backup_20260329_214945/file_counter.sh
 create mode 100644 backup_20260329_214945/system_info.sh
 create mode 100644 calculator.sh
 create mode 100644 check_file.sh
 create mode 100644 file_counter.sh
 create mode 100644 system_info.sh
```

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